Example 1.12

Let $A = \{x : x$ is an even natural number and $1 < x \leq 12\}$ and $B = \{x : x$ is a multiple of 3, $x \in \mathbb{N}$ and $x \leq 12\}$ be two sets. Find $A \cap B$.

Solution

Here $A = \{2, 4, 6, 8, 10, 12\}$ and $B = \{3, 6, 9, 12\}$

$A \cap B = \{6, 12\}$

Example 1.13

If $A = \{2, 3\}$ and $C = \{\ \}$, find $A \cap C$.

Solution

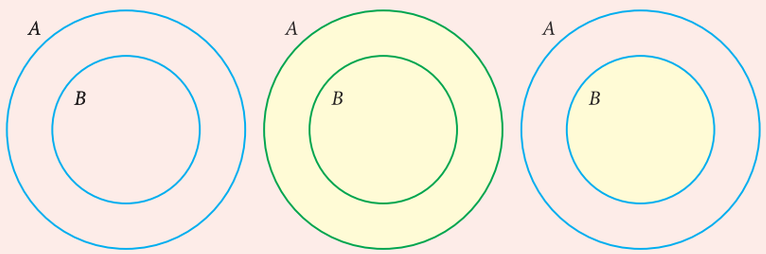
There is no common element and hence $A \cap C = \{\ \}$

Note

- $A \cap A = A$ (ii) $A \cap \varnothing = \varnothing$
- $A \cap U = A$ where A is any subset of universal set U
- $A \cap B \subseteq A$ and $A \cap B \subseteq B$
- $A \cap B = B \cap A$ (Intersection of two sets is commutative)

Note

- When $B \subset A$, the union and intersection of A and B are represented in Venn diagram as follows



$B \subset A$ | shaded region is $A \cup B = A$ | shaded region is $A \cap B = B$

Fig. 1.10 | Fig. 1.11 | Fig. 1.12

- If A and B are any two non empty sets such that $A \cup B = A \cap B$, then $A = B$
- Let $n(A) = p$ and $n(B) = q$ then
  - (a) Minimum of $n(A \cup B) = \max\{p, q\}$
  - (b) Maximum of $n(A \cup B) = p + q$
  - (c) Minimum of $n(A \cap B) = 0$
  - (d) Maximum of $n(A \cap B) = \min\{p, q\}$

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